

JAVA INHERITANCE PROGRAMS

1. Single Inheritance

Code:

```
package inheritance;

class Ones {
    int x = 27;

    void show() {
        System.out.println("this is one");
    }
}

class Twos extends Ones {
    int y = 34;

    void see() {
        System.out.println("see :" + super.x + " " + this.y);
    }
}

public class Single {
    public static void main(String[] args) {
        Twos t = new Twos();
        t.see();
        t.show();
    }
}
```

Output:

Command Prompt

```
C:\Users\Student>javac Single.java

C:\Users\Student>java inheritance.Single
see :27 34
this is one

C:\Users\Student>
```

2. Multilevel Inheritance

Code:

```
package inheritance;

class C1 {
    int x = 27;

    void show() {
        System.out.println("this is one");
    }
}

class C2 extends C1 {
    int y = 34;

    void see() {
        System.out.println("see :" + super.x + " " + this.y);
    }
}

class C3 extends C2 {
    int z = 12;

    void demo() {
        System.out.println("see :" + super.x + " " + super.y + " " + this.z);
    }

    void addition() {
        System.out.println("addition :" + (x + y + z));
    }
}

public class Multilevel {
    public static void main(String[] args) {
        C3 T = new C3();
        T.demo();
        T.addition();
    }
}
```

Output:

Command Prompt

```
C:\Users\Student>javac Multilevel.java

C:\Users\Student>java inheritance.Multilevel
see :27 34 12
addition :73

C:\Users\Student>
```

3. Hierarchical Inheritance

Code:

```
package inheritance;

class one {
    int x = 27;

    void show() {
        System.out.println("this is one");
    }
}

class two extends one {
    int y = 34;

    void see() {
        System.out.println("see :" + super.x + " " + this.y);
    }
}

class three extends one {
    int z = 12;

    void demo() {
        System.out.println("see :" + super.x + " " + this.z);
    }
}

public class Hierarchical {
    public static void main(String[] args) {
        two w = new two();
        three r = new three();

        w.see();
        r.demo();
    }
}
```

Output:

Command Prompt

```
C:\Users\Student>javac Hierarchical.java

C:\Users\Student>java inheritance.Hierarchical
see :27 34
see :27 12

C:\Users\Student>
```

4. Multiple Inheritance (Using Interfaces)

Code:

```
package inheritance;

interface riya {
    int x = 10;

    void siva();
}

interface raji {
    int z = 15;

    void lucky();
}

class Raghu implements riya, raji {
    @Override
    public void siva() {
        System.out.println("i am siva");
    }

    @Override
    public void lucky() {
        System.out.println("i am lucky");
    }
}

public class Inter {
    public static void main(String[] args) {
        Raghu r = new Raghu();
        r.siva();
        r.lucky();
    }
}
```

Output:

Command Prompt

```
C:\Users\Student>javac Inter.java

C:\Users\Student>java inheritance.Inter
i am siva
i am lucky

C:\Users\Student>
```

5. Hybrid Inheritance

Code:

```
package inheritance;

class Cls1 {
    int x = 27;

    void show() {
        System.out.println("this is one");
    }
}

class Cls2 extends Cls1 {
    int y = 34;

    void see() {
        System.out.println("see :" + super.x + " " + this.y);
    }
}

class Cls3 extends Cls2 {
    int z = 12;

    void demo() {
        System.out.println("see :" + super.x + " " + super.y + " " + this.z);
    }
}

class Cls4 extends Cls1 {
    int z = 12;

    void demos() {
        System.out.println("see :" + super.x + " " + this.z);
    }
}

public class Hybrid {
    public static void main(String[] args) {
        Cls3 c = new Cls3();
        c.demo();

        Cls4 s = new Cls4();
        s.demos();
    }
}
```

Output:

Command Prompt

```
C:\Users\Student>javac Hybrid.java

C:\Users\Student>java inheritance.Hybrid
see :27 34 12
see :27 12

C:\Users\Student>
```